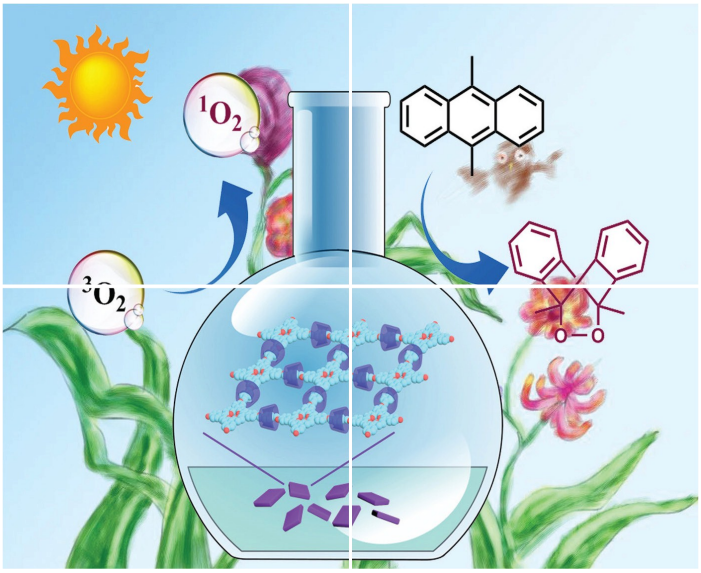
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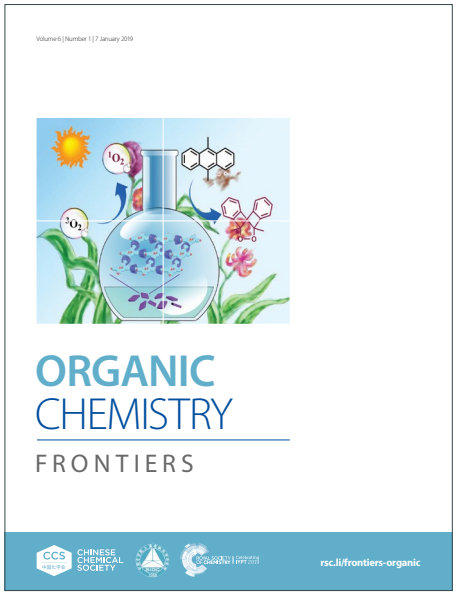
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ARTICLE

Catalyst-Controlled Formal [4+1] Annulation of *N*-Vinyl Fluorenone Nitrones and Allenoates to Prepare Spirofluorenylpyrrolines

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

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We report a readily commercial Gimeracil-catalyzed formal [4+1] annulation approach for the synthesis of spirofluorenylpyrrolines in good yields with high diastereoselectivity from easily available *N*-vinyl fluorenone nitrones and allenoates. The reaction showed broad substrate scope and tolerated a variety of functional groups on both *N*-vinyl fluorenone nitrones and allenoates. Mechanistic studies showed that the catalyst might play important roles on promoting enolation and selective cyclization of 1,3-dicarbonyl intermediates. More importantly, the obtained spirofluorenylpyrrolines were easily converted to various spirofluorenylpyrroline-based scaffolds.

Introduction

The spirocyclic compounds are currently of great interest because 3D structures of spirocyclic scaffolds play important roles on the biological and pharmacological activities.[1] The construction of spirocyclic scaffolds has long been considered significant but is a highly challenging task owing to its inherent conformational rigidity and steric hindrance.[2,3] Therefore, development of new methods toward the preparation of spirocyclic compounds is desirable. Cycloaddition reaction, among the most efficient annulations to build heterocycles, has been successfully used to construct spirocyclic compounds.[4] Nitrones have been extensively used as 1,3-dipolar compounds in cycloaddition reactions to construct N, O-heterocycles since its discovery, which can not only participate in [3+2], [3+3], [4+3] or other relative cycloadditions, but also be used as *O*-sources in the *O*-transfer reaction.[5,6] The [3+2] cycloaddition of nitrones with allenoates was reported before but it was always used to prepare 2,3-disubstituted indoles or related benzoazepine scaffords (Scheme 1-A).[7,8] In 2019, Prathapan and coworkers, developed a catalyst-free synthesis of 1-azaspiro[4.4]nonane-3-one through a [3+2] cycloaddition and sequential [1,3]-rearrangement of 9-fluorenone nitrones and 1,1-disubstitued allenes, however, the substrate scope was limited (Scheme 1-B).[9] Recently, *N*-vinyl nitrones have been used as useful organic building blocks because the double bond of nitrone *N*-atom can participate in many transformations to biuld complex targets.[10] However, the cycloaddition of *N*-vinyl nitrones with allenoates was only one example reported up to now. In 2014, Anderson and co-workers found that *N*-vinyl fluorenone nitrones reacting with electron-deficient allenes and sequential [3,3]-rearrangement afforded various 1,4-enamino ketones under catalyst-free conditions, which then underwent hydrolysis and condensation to result in pyrrole target (Scheme 1-C).[11] They proposed a simple 1,4-addition reaction mechanism rather than undergoing [3+2] cycloaddition due to the sterically hindered and electronically deactivated fluorenyl imine group. These afforementioned reactions have showed their efficiency of nitrones to prepare indole or pyrrole heterocycles under the mild reaction coniditions, but they usually focused on a [3+2] annulation and the allenes served as 2C-or 3C-synthons. On the other hand, these reported examples have showed that the *O*-atom of nitrones was usually not transferred to the target products and ketones were always produced as waste after the reaction completed. Moreover, spirocyclic compounds containing fluorenyl groups usually possessed distinct properties owing to their high rigidity and steric hindrance.[12] Therefore, the development of atom economic strategies to construct spirocyclic compounds containing fluorenyl groups is also valuable.

Recently, we have successfully developed [3+2] cycloaddition of 9-fluorenone nitrones with methylenecyclopropanes to afford various spirofluorenyl-β-lactams and spiropiperidin-4-ones, and [3+3] cycloaddition with aza-oxyallyl cations to afford spirofluorenyl-1,2,4-oxadiazinan-5-ones.[13] During our studies on the applications of *N*-vinyl nitrones,[14] compared to Anderson's proposed mechanism,[11] we supposed that a hydrogen-bonding catalyst might promote the 1,4-nucleophilic addition of *N*-vinyl fluorenone nitrones to allenoates and

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Electronic Supplementary Information (ESI) available: Experimental details and characterization of all compounds and copies of ^{1}H and ^{13}C NMR for new compounds. CCDC 1974919. For ESI and crystallographic data in CIF. See DOI: 10.1039/x0xx00000x

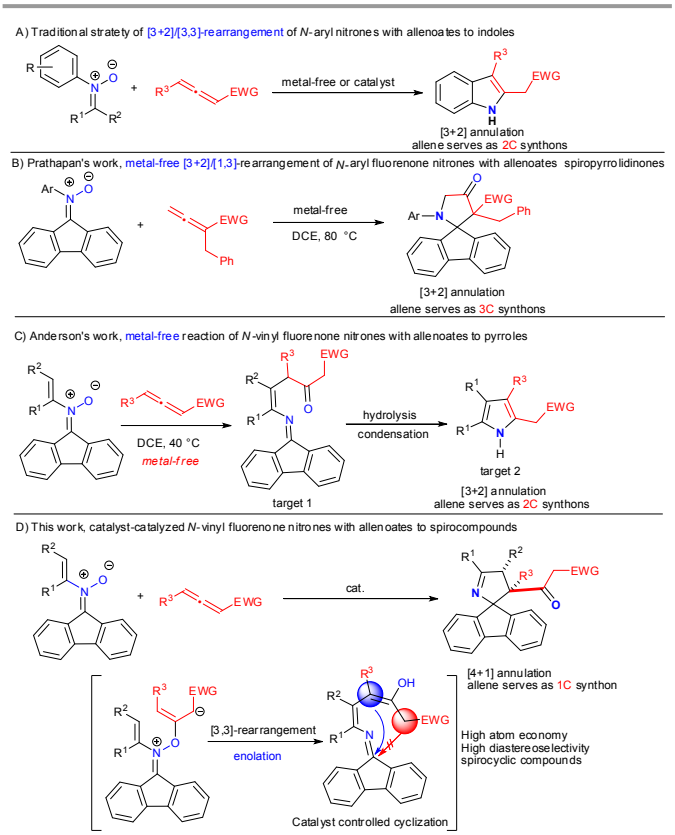
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sequential selective enolation and cyclization to avoid hydrolysis affording spirofluorenylpyrrolines (Scheme 1-C). Thus, the allenes would serve as 1C-synthon. Herein, we developed a metal-free and simple Gimeracil-controlled formal [4+1] annulation of *N*-vinyl fluorenone nitrones with allenoates to afford functionalized spirofluorenylpyrrolines with high atom economy and diastereoselectivity.



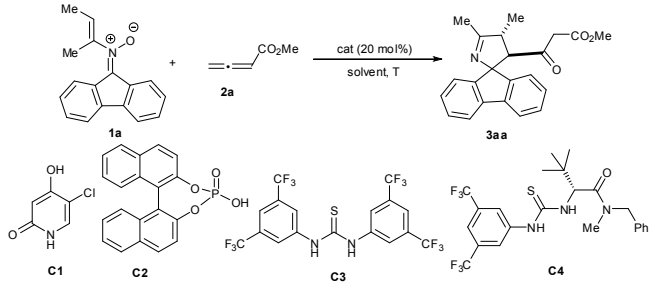
Scheme 1. New design for the [4+1] annulation of nitrones and allenoates

Results and Discussion

Initially, *N*-vinyl fluorenone nitrone **1a** and allenoate **2a** were chosen as model substrates to test the effect of catalyst on the formal [4+1] annulation. As shown in Table 1, when **1a** was reacted with **2a** without catalyst in DCE at 40 °C, desired spirofluorenylpyrroline **3aa** was not observed (Table 1, entry 1). Pleasingly, 20% yield of **3aa** was obtained when 20 mol% of Gimeracil (**C1**) was added to the reaction mixture (Table 1, entry 2). It is worthy to note that adding Gimeracil catalyst obviously shortened the reaction time to 12 h compared to Anderson's work[10] with 18-24 h, suggesting that Gimeracil could accelerate the reaction. Increasing the reaction temperature to 80 °C improved the yield of **3aa** to 48% but decreased to 12% at 100 °C (Table 1, entries 3-4). We surmised that the hydrogen-bonding catalysts might promote the reaction, and then, phosphine catalyst **C2** and thiourea catalysts **C3-C4** were also tested. We found that **C2** catalyst inhibited the reaction while **C3** catalyst afforded **3aa** in 34% yield (Table 1, entries 5-6). Adding chiral thiourea **C4** furnished **3aa** in 31% yield without enantioselectivity (Table 1, entry 7).

The ammonium chloride (NH_4Cl) was also tested as catalyst and found that **3aa** was afforded only in 8% yield (Table 1, entry 8). Solvent screening revealed that lower yields of **3aa** were obtained in toluene, $CHCl_3$, C_6F_6, and MeCN (Table 1, entries 9-12). We are pleased to find that DMF as solvent delivered **3aa** in 66% yield (Table 1, entry 13). Decreasing the amount of Gimeracil to 10 mol% afforded **3aa** in 51% yield (Table 1, entry 14). Increasing the amount of Gimeracil to 30 mol% improved the yield of **3aa** to 69% and further to 77% by adding 50 mol% of Gimeracil (Table 1, entries 15-16). Therefore, the optimal conditions for preparing spirofluorenylpyrroline **3aa** was Gimeracil (50 mol%) in DMF at 80 °C for 3 h.[15]

Table 1. Optimization of the reaction conditions.[a]



entry	cat.	solvent	T (°C)	time (h)	**3aa**,yield %[b]
1	-	DCE	40	24	<5
2	**C1**	DCE	40	12	20
3	**C1**	DCE	80	12	48
4	**C1**	DCE	100	6	12
5	**C2**	DCE	80	24	0
6	**C3**	DCE	80	24	34
7	**C4**	DCE	80	24	31[c]
8	NH_4Cl	DCE	80	24	8
9	**C1**	toluene	80	24	<10
10	**C1**	$CHCl_3$	80	24	39
11	**C1**	C_6F_6	80	24	37
12	**C1**	MeCN	80	24	29
13	**C1**	DMF	80	8	66
14	**C1**	DMF	80	24	51[d]
15	**C1**	DMF	80	6	69[e]
16	**C1**	DMF	80	3	77[f]

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol, 2.0 equiv.), cat. (20 mol%), solvent (2.0 mL); [b] isolated yield; [c] ee <5% [d] **C1** (10 mol%); [e] **C1** (30 mol%); [f] **C1** (50 mol%).

With the optimized conditions in hand, we start to explore the substrate scope of *N*-vinyl fluorenone nitrones **1** with **2a** for the preparation of spirofluorenylpyrrolines under the Gimeracil-catalysis conditions. As shown in Table 2, it was found that all reactions proceeded smoothly to afford the corresponding spirofluorenylpyrrolines **3aa-3la** in good yields ranging from 32% to 77% with high diastereoselectivity. The diastereoselectivity of these compounds were determined by ^{1}H NMR spectra of the crude mixtures. The *N*-substituent of fluorenone nitrones tolerated a variety of vinyl groups on the *N*-atom, such as linear mono- or disubstituted, and cyclic vinyl groups. The cyclic vinyl groups could be present with five to seven-membered rings and resulted in **3fa-3ja** in good yields. Interestingly, *N*-substituent of product **3ia** containing a pyran

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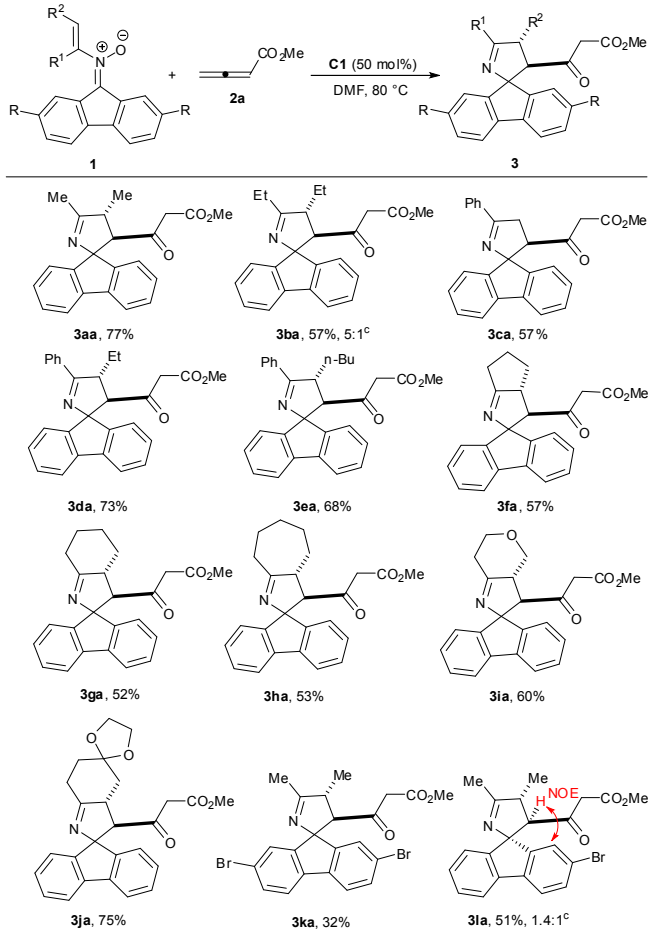
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ring was obtained in 60% yield. Fluorenone nitrone **1j** with an acetal group on the six-membered ring was compatible with the reaction conditions to afford **3ja** in 75% yield. The acetal group in compound **3ja** might allow more transformations of these compounds by further deprotection. When the fluorenyl groups of nitrone **1k** with two bromo groups and **1l** with one bromo group were carried out under the optimized conditions, products **3ka** and **3la** were obtained in 32% and 51% yields, respectively. However, compound **3la** was afforded with a 1.4:1 diastereoselectivity on the fluorenyl group and the major isomer showed that proton at the α-position of ketone had correlations with proton of the aryl ring.

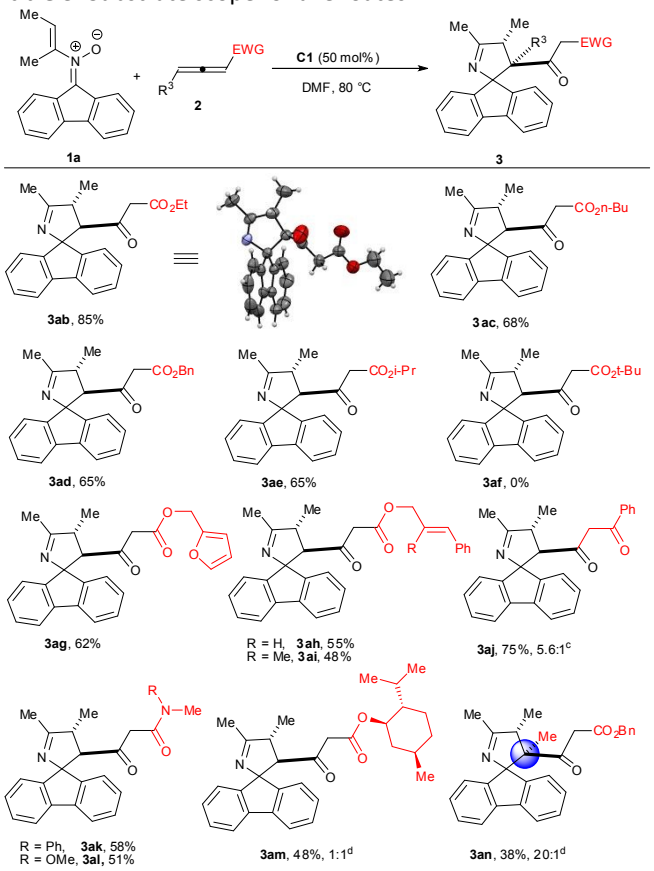
Table 2. Substrate scope for *N*-vinyl fluorenone nitrones **1.** [a,b]



[a] Reaction conditions: **1** (0.2 mmol), **2a** (0.4 mmol, 2.0 equiv.), **C1** (50 mol%), DMF (2.0 mL), 80 °C, 3–6 h; [b] isolated yield; [c] *dr* value.

Next, various allenoates **2** were evaluated to determine their effect on the preparation of spirofluorenylpyrrolines. As shown in Table 3, all the allenoates **2b-2n** were reacted with nitrone **1a** smoothly to afford the corresponding products **3ab-3an** in moderate to good yields with *trans* diastereomers only. However, when *t*-Bu substituted allenoate **2f** was carried out under the optimized conditions, compound **3af** was not observed perhaps due to the large steric hindrance of allenoate. The allenoates tolerated various electron-withdrawing groups (EWG), such as ester, ketone, and amides, delivering the corresponding spirofluorenylpyrrolines in good yields. It is worthy to note that compound **3aj** was obtained as a mixture of enol and ketone with a 5.6:1 ratio owing to the existing enolation of 1,3-diketone group. The structure of compound **3** was determined by X-ray diffraction analysis of **3ab**.[16] When reaction with (-)-menthol-derived allenoate **2m** was carried out under the optimized conditions, product **3am** was obtained in 48% yield with a 1:1 diastereoselectivity, showing that the auxiliary group of allenoates cannot control the diastereoselectivity of the pyrroline ring. Interestingly, methyl disubstituted allenoate **2n** was subjected to the optimized conditions affording product **3an** containing an all quaternary carbon center in 38% yield with a 20:1 *dr*. The relative configuration of **3an** was determined by its NOESY spectrum, showing that the two methyl groups were in the same face.

Table 3. Substrate scope for allenoates **2**.[a,b]



[a] Reaction conditions: **1a** (0.2 mmol), **2** (0.4 mmol, 2.0 equiv.), **C1** (50 mol%), DMF (2.0 mL), 80 °C, 3–6 h; [b] isolated yield; [c] the ratio of enol and ketone; [d] *dr* value.

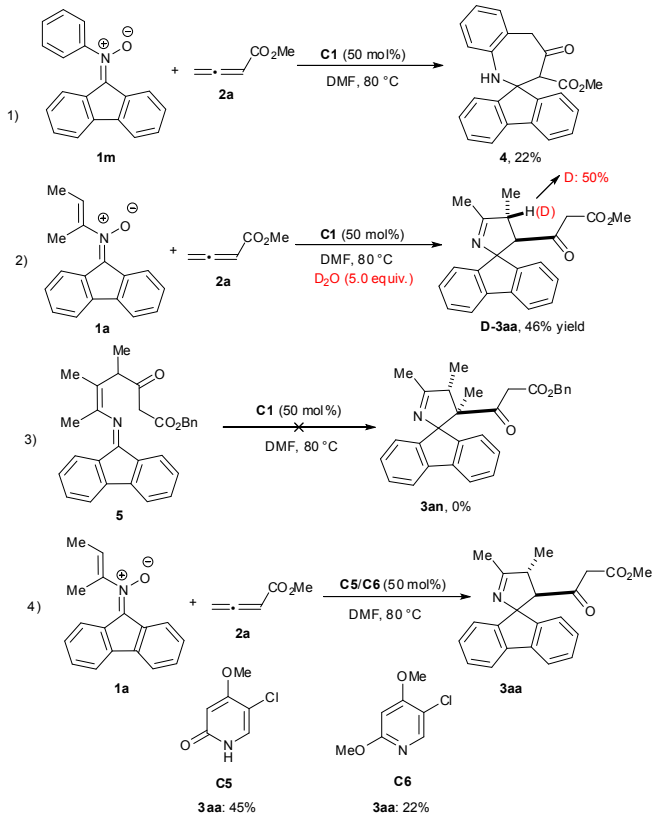
When *N*-phenyl fluorenone nitrone **1m** was reacted with allenoate **2a** under the standard conditions, spirobenzoazepine **4** was obtained in 22% yield accompanied by 9-fluorenone and indole by-products (Scheme 2-1). This reaction showed an obvious different selectivity compared to Prathapan's work in Scheme 1-B, and the spirobenzoazepine **4** might be formed by intramolecular cyclization at the last step of the whole cascade reaction. To better understand the mechanism for the formation of spirofluorenylpyrroline **3**,

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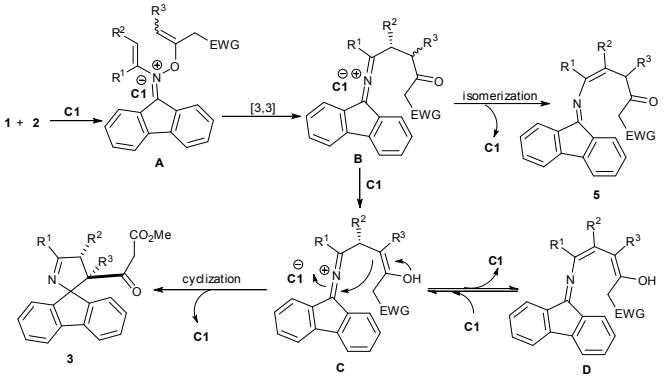
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control experiments were performed. Nitrone **1a** reacting with **2a** in the presence of 5.0 equiv. of D_2O affording product **D-3aa** in 46% yield with a 50% deuterated ratio at the α-position of imine group of pyrroline (Scheme 2-2). These results indicated an isomerisation between imine and enamine. When compound **5** prepared by Anderson's method[11] was carried out under our optimized conditions, desired spirofluorenylpyrroline **3an** was not observed and compound **5** was recovered accompanied by decomposition to 9-fluorenone (Scheme 2-3). These results revealed that compound **5** might be not a real intermediate for the formation of compound **3an**. The role of Gimeracil was also investigated, as shown in Scheme 2-4, using **C5** and **C6** catalysts gave **3aa** only in 45% and 22% yields, respectively. These results showed that the protons of Gimeracil (**C1**) had great impact on the reaction yields, which might play as acidity to promote the enolation and hydrogen-bonding to activate the dicarbonyl group owing to the catalyst's resonance between dihydropyrimidine and 2-pyridinone.



Scheme 2. Mechanistic studies

Based on the above experimental results, a plausible mechanism for the formation of spirofluorenylpyrrolines **3** from *N*-vinyl fluorenone nitrones **1** and allenoates **2** is illustrated in Scheme 3. The initial 1,4-nucleophilic addition of nitrone **1** and allenoate **2** under Gimeracil catalyst affords intermediate **A**,[17] which undergoes [3,3]-rearrangement to result in intermediate **B**. The dicarbonyl groups of Intermediate **B** go through a selective enolation under the acidic **C1**-catalyst to furnish four-substituted enol intermediate **C**. Isomerisation of **B** delivers compound **5**. Also, intermediate **C** isomerizes to intermediate **D** which exists a balance with intermediate **C** under the **C1** catalyst. Finally, kinetically controlled intramolecular cyclization of intermediate **C** from the opposite face with R^2 group provides product **3** and releases **C1** catalyst. Under our optimized conditions, we does not observe compound **5**, suggesting that **C1** catalyst might play the roles of acidity and hydrogen-bonding, which accelerate the reaction rate and facilitate the conversion of intermediate **B** to intermediate **C** instead of isomerisation to compound **5**. The balance between **C** and **D** was supported by the experimental results of Scheme 2-2.



Scheme 3. Proposed mechanism

To demonstrate the advantages of this formal [4+1] annulation strategy, a gram scale performance of nitrone **1a** was carried out under the optimal conditions (Scheme 4-1). When 1.0 g of *N*-vinyl fluorenone nitrone **1a** was reacted with **2a**, product **3aa** was obtained in 65% yield (0.89 g). Then, the transformations of **3aa** were investigated (Scheme 4-2). Treating **3aa** with $NaBH_4$ in MeOH at room temperature afforded amino alcohol **6** in 61% yield with only one diastereomer, showing that the ketone, ester and imine groups were all reduced. Compound **3aa** reacting with cyclopropenone under DBU in Et_2O delivered spirofluorenylpyrrole-tethered γ-lactone **7** in 55% yield with a 1.8:1 *dr*. Reaction of **3aa** with benzene-1,2-diamine afforded compound **8** in 35% yield. These easy transformations expand the applications in the diversity of spirofluorenyl compounds and will allow more potent biological activity studies.

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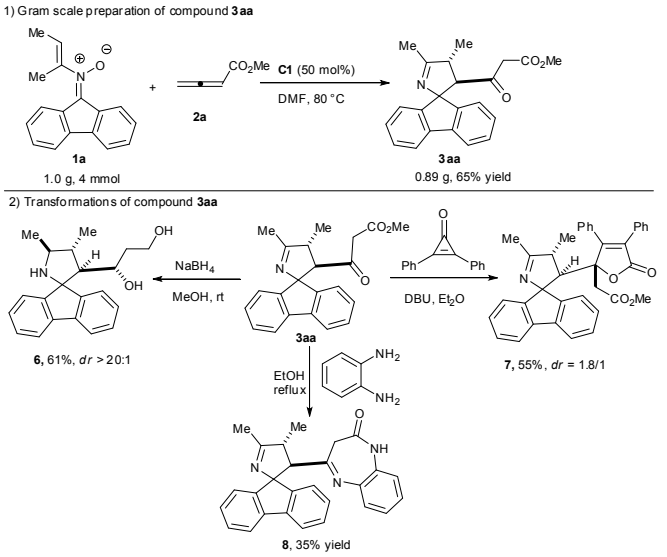
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Scheme 4. Gram scale preparation of **3aa** and its applications

Conclusions

We have developed a metal-free atom economic strategy through a simple Gimeracil-controlled formal [4+1] annulation of *N*-vinyl fluorenone nitrones with allenoates. This reaction tolerates a broad range of *N*-vinyl fluorenone nitrones and allenoates to furnish various densely functionalized spirofluorenylpyrrolines. The present method features allenes as 1C-synthon and Gimeracil as a catalyst to achieve an atom economy reaction to afford spirocyclic compounds. More importantly, we have devised a novel platform for the *N*-vinyl fluorenone nitrones with allenoates, and it will provide a further design for other dipolarophiles into new reaction mode.

Conflicts of interest

There are no conflicts interest to declare.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21871062), the Natural Science Foundation of Guangxi (2016GXNSFFA380005), the State Key Laboratory for Chemistry and Molecular Engineering of Medicinal Resources (CMEMR2017-A01), the Ministry of Education of China (IRT_16R15), the "Overseas 100 Talents Program" of Guangxi Higher Education, and the "One Thousand Young and Middle-Aged College and University Backbone Teachers Cultivation Program" of Guangxi are greatly appreciated.

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15 Other substituted groups of Gimeracil catalysts were also investigated. Please see more details in Supporting Information.

16 CCDC: 1974919 (**3ab**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge

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17 The mechanism was proposed according to Anderson's work in reference 11. However, intermediate **A** might be more possible because of 50 mol% of acidic Gimeracil catalyst in the reaction. The acidic conditions might facilitate enolation to form intermediate **C**.

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